

11. Illustrate the concept of multithreading using a C program.

AIM:

To write and execute a C program to illustrate the concept of **multithreading** using POSIX threads.

CODE:

```
#include <stdio.h>

#include <pthread.h>

void *thread1(void *arg)
{
    printf("Thread 1 is running\n");
    for (int i = 1; i <= 5; i++)
        printf("Thread 1: %d\n", i);

    return NULL;
}

void *thread2(void *arg)
{
    printf("Thread 2 is running\n");
    for (int i = 1; i <= 5; i++)
        printf("Thread 2: %d\n", i);

    return NULL;
}

int main()
{
    pthread_t t1, t2;
```

```

pthread_create(&t1, NULL, thread1, NULL);

pthread_create(&t2, NULL, thread2, NULL);


pthread_join(t1, NULL);

pthread_join(t2, NULL);


printf("Both threads completed.\n");


return 0;
}

```

OUTPUT:

main.c	Run	Output
<pre> 1 #include <stdio.h> 2 #include <pthread.h> 3 4 void *thread1(void *arg) 5 { 6 printf("Thread 1 is running\n"); 7 for (int i = 1; i <= 5; i++) 8 printf("Thread 1: %d\n", i); 9 10 return NULL; 11 } 12 13 void *thread2(void *arg) 14 { 15 printf("Thread 2 is running\n"); 16 for (int i = 1; i <= 5; i++) 17 printf("Thread 2: %d\n", i); 18 19 return NULL; 20 } 21 22 int main() 23 { 24 pthread_t t1, t2; 25 26 pthread_create(&t1, NULL, thread1, NULL); 27 pthread_create(&t2, NULL, thread2, NULL); 28 29 pthread_join(t1, NULL); 30 pthread_join(t2, NULL); 31 32 printf("Both threads completed.\n"); 33 34 return 0; 35 } </pre>		<pre> Thread 1 is running Thread 1: 1 Thread 1: 2 Thread 1: 3 Thread 1: 4 Thread 2 is running Thread 2: 1 Thread 2: 2 Thread 2: 3 Thread 2: 4 Thread 2: 5 Thread 1: 5 Both threads completed. --- Code Execution Successful --- </pre>

RESULT

Thus, the C program was successfully executed to demonstrate **multithreading**, where two threads perform their tasks concurrently.